

Non-uniform Non-Archimedean Estimate for Regular Domains

Statement

Let $(X, \mu, \mathcal{P})$ be a good grid and let

$$0 < s < \frac{1}{p}, \quad 1 \leq p < \infty, 1 \leq q \leq \text{infity}$$

For $Q \in \mathcal{P}^k$, write $|Q| = \mu(Q)$, and use the Souza atom

$$\psi_Q = |Q|^{s-\frac{1}{p}} \mathbf{1}_Q.$$

There is $C_{na} \geq 0$ such that the following holds: Let $f \in \mathcal{B}_{p,q}^s \cap L^\infty$ be represented

$$f = \sum_{k \geq 0} \sum_{Q \in \mathcal{P}^k} c_Q \psi_Q,$$

by either an standard representation or a positive representation, with coefficient finite cost

$$\|c\|_{b_{p,q}^s} = \begin{cases} \left(\sum_{k \geq 0} \left(\sum_{Q \in \mathcal{P}^k} |c_Q|^p \right)^{q/p} \right)^{1/q}, & q < \infty, \\ \sup_{k \geq 0} \left(\sum_{Q \in \mathcal{P}^k} |c_Q|^p \right)^{1/p}, & q = \infty. \end{cases}$$

Assume also that

$$|f(x)| \leq M \quad \text{for a.e. } x.$$

Let $(\Omega_i)_{i \in \Lambda}$ be pairwise disjoint domains, such that $\mathbf{1}_{\Omega_i}$ with a positive representation

$$\mathbf{1}_{\Omega_i} = \sum_{k \geq 0} \sum_{Q \in \mathcal{P}^k} b_Q^i \psi_Q$$

such that

$$b_Q^i \neq 0 \implies b_Q^i > 0 \implies Q \subset \Omega_i, \quad |b_Q^i| \leq \mathcal{C}_i,$$

and such that $\|c\|_{b_{p,q}^s} \leq \mathcal{C}_i$

Let $(\Theta_i)_{i \in \Lambda}$ be real weights. The main hypothesis is that, for every active source cell P , that is, for every $P \in \mathcal{P}$ with $c_P \neq 0$,

$$\sum_{\substack{i \in \Lambda \\ P \cap \Omega_i \neq \emptyset}} |\Theta_i| (1 + \mathcal{C}_i) \leq N. \quad (1)$$

Define

$$h = \sum_{i \in \Lambda} \Theta_i \mathbf{1}_{\Omega_i} f.$$

Then there is a representation S of h , supported in the domains Ω_i , such that

$$\text{pqCost}(S) + \|h\|_{L^\infty} \leq C_{\text{na}} N (\text{pqCost}(R) + M), \quad (2)$$

where $R = (c_Q)_Q$ is the chosen representation of f , and C_{na} depends only on the structural constants of the grid and on s, p, q . Moreover, if a coefficient of S at a cell Q is nonzero, then

$$Q \subset \Omega_i \quad \text{for some } i \in \Lambda.$$

The Product Decomposition

Suppose first that Λ is finite. If

$$\mathbf{1}_{\Omega_i} = \sum_Q b_Q^i \psi_Q \quad \text{and} \quad f = \sum_P c_P \psi_P.$$

The product is decomposed into two parts:

$$\left(\sum_{i \in \Lambda} \Theta_i \mathbf{1}_{\Omega_i} \right) f = \sum_{i \in \Lambda} U_{1,i} + \sum_{i \in \Lambda} U_{2,i}.$$

The first term $U_{1,i}$ collects the interactions where the indicator cell is below the source cell. If $Q \in \mathcal{P}^j$, its coefficient is

$$(U_{1,i})_Q = \Theta_i b_Q^i \mathcal{T}_R(Q), \quad (3)$$

where

$$\mathcal{T}_R(Q) = \sum_{\ell \leq j} \sum_{\substack{P \in \mathcal{P}^\ell \\ Q \subset P}} c_P |P|^{s - \frac{1}{p}}. \quad (4)$$

Since the representation of f is either standard or positive we have

$$\|\mathcal{T}_R(Q)\| \leq M.$$

This is the ancestral tower of the source representation R , evaluated on the cell Q . In Lean this is

$$\text{weightedAncestorCoeffSum}(G, R, Q).$$

The second term $U_{2,i}$ collects the complementary interactions, where the source cell is below the indicator cell. Its coefficients have the schematic form

$$(U_{2,i})_J = \Theta_i c_J \mathcal{T}_i^<(J), \quad (5)$$

where $\mathcal{T}_i^<(J)$ is the strict ancestor tower of the indicator representation above J . More explicitly, if $J \in \mathcal{P}^j$, then

$$\mathcal{T}_i^<(J) = \sum_{\ell < j} \sum_{\substack{Q \in \mathcal{P}^\ell \\ J \subset Q}} b_Q^i |Q|^{s - \frac{1}{p}}. \quad (6)$$

The word “strict” means that the level $\ell = j$ is excluded. Since this representation is positive we have

$$\|\mathcal{T}_i^<(J)\| \leq \|\mathbf{1}_{\Omega_i}\|_\infty \leq 1.$$

In the Lean formalization this is

$$\text{strictWeightedAncestorCoeffSum}(G, R_i, J),$$

where R_i is the chosen indicator representation of $\mathbf{1}_{\Omega_i}$.

Define

$$U_1 = \sum_{i \in \Lambda} U_{1,i}$$

and

$$U_2 = \sum_{i \in \Lambda} U_{2,i}$$

We have

$$(U_1)_Q = \sum_{i \in \Lambda} \Theta_i b_Q^i \mathcal{T}_R(Q),$$

$$(U_2)_J = \sum_{i \in \Lambda} \Theta_i c_J \mathcal{T}_i^<(J),$$

The U_1 estimate

The cost at level k of U_1 is, by Minkovsky

$$\left(\sum_{Q \in \mathcal{P}^k} \left\| \sum_{i \in \Lambda} \Theta_i b_Q^i \mathcal{T}_R(Q) \right\|^p \right)^{1/p} \quad (7)$$

$$\leq \left(\sum_{Q \in \mathcal{P}^k} \left\| \sum_{i \in \Lambda} \Theta_i b_Q^i \mathcal{T}_R(Q) \right\|^p \right)^{1/p} \quad (8)$$

$$\leq M \left(\sum_{Q \in \mathcal{P}^k} \left\| \sum_{i \in \Lambda} \Theta_i b_Q^i \right\|^p \right)^{1/p} \quad (9)$$

$$\leq M \sum_{i \in \Lambda} \|\Theta_i\| \left(\sum_{Q \in \mathcal{P}^k} \|b_Q^i\|^p \right)^{1/p} \quad (10)$$

so

$$\left(\sum_k \left(\sum_{Q \in \mathcal{P}^k} \left\| \sum_{i \in \Lambda} b_Q^i \mathcal{T}_R(Q) \right\|^p \right)^{q/p} \right)^{1/q} \quad (11)$$

$$\leq M \left(\sum_k \left(\sum_{i \in \Lambda} \|\Theta_i\| \left(\sum_{Q \in \mathcal{P}^k} \|b_Q^i\|^p \right)^{1/p} \right)^q \right)^{1/q} \quad (12)$$

$$\leq M \sum_{i \in \Lambda} \|\Theta_i\| \left(\sum_k \left(\sum_{Q \in \mathcal{P}^k} \|b_Q^i\|^p \right)^{q/p} \right)^{1/q} \quad (13)$$

$$\leq M \sum_{i \in \Lambda} \|\Theta_i\| \mathcal{C}_i. \quad (14)$$

For a fixed $Q \in \mathcal{P}^j$, suppose

$$\Theta_i (U_{1,i})_Q \neq 0.$$

Then $b_Q^i \neq 0$, hence $Q \subset \Omega_i$. Also, if $\mathcal{T}_R(Q) \neq 0$, then some ancestor $P \supset Q$ has $c_P \neq 0$. Thus $P \cap \Omega_i \neq \emptyset$. Applying the overlap hypothesis (1) to this active cell P gives

$$|\Theta_i| (1 + \mathcal{C}_i) \leq N.$$

Since $|b_Q^i| \leq \mathcal{C}_i \leq 1 + \mathcal{C}_i$, we obtain the local bound

$$|\Theta_i| |b_Q^i| \leq N. \quad (15)$$

Using (3),

$$|\Theta_i(U_{1,i})_Q| = |\Theta_i| |b_Q^i| |\mathcal{T}_R(Q)| \leq N |\mathcal{T}_R(Q)|.$$

Now use disjointness of the domains. For a fixed Q , at most one index i can have $b_Q^i \neq 0$, because $b_Q^i \neq 0$ implies $Q \subset \Omega_i$, and the Ω_i 's are pairwise disjoint. Therefore, for every finite $\Gamma \subset \Lambda$,

$$\begin{aligned} \sum_{i \in \Gamma} |\Theta_i|^p \sum_{Q \in \mathcal{P}^j} |(U_{1,i})_Q|^p &= \sum_{Q \in \mathcal{P}^j} \sum_{i \in \Gamma} |\Theta_i(U_{1,i})_Q|^p \\ &\leq \sum_{Q \in \mathcal{P}^j} N^p |\mathcal{T}_R(Q)|^p \\ &= N^p T_j(R)^p. \end{aligned} \quad (16)$$

The U_2 estimate

The U_2 term is controlled by the source coefficients themselves. Indeed, the strict indicator tower $\mathcal{T}_i^<(J)$ is nonzero only when an ancestor of J belongs to the canonical decomposition of Ω_i . Thus, if $\Theta_i(U_{2,i})_J \neq 0$, then some source cell meeting the corresponding indicator ancestor also meets Ω_i , and the overlap hypothesis again gives

$$|\Theta_i| (1 + \mathcal{C}_i) \leq N.$$

The canonical indicator construction gives

$$|\mathcal{T}_i^<(J)| \leq 1 + \mathcal{C}_i.$$

Hence

$$|\Theta_i(U_{2,i})_J| \leq N |c_J|.$$

As before, disjointness implies that for each J at most one i may contribute. Consequently,

$$\begin{aligned} \sum_{i \in \Gamma} |\Theta_i|^p \sum_{J \in \mathcal{P}^j} |(U_{2,i})_J|^p &\leq N^p \sum_{J \in \mathcal{P}^j} |c_J|^p \\ &= N^p A_j(R)^p. \end{aligned} \quad (17)$$

Combining U_1 and U_2

The product representation for the finite partial sum

$$h_\Gamma = \sum_{i \in \Gamma} \Theta_i \mathbf{1}_{\Omega_i} f$$

has level j coefficient power bounded by the weighted sum of $U_{1,i} + U_{2,i}$. Using

$$|x + y|^p \leq 2^{p-1} (|x|^p + |y|^p),$$

together with (16) and (17), we get

$$\sum_{i \in \Gamma} |\Theta_i|^p \sum_{Q \in \mathcal{P}^j} |(U_{1,i} + U_{2,i})_Q|^p \leq 2^{p-1} N^p (T_j(R)^p + A_j(R)^p). \quad (18)$$

This is the non-uniform level-by-level estimate. Notice that the factor $|\Theta_i| \mathcal{C}_i$ is used locally before summing in Q ; it is not replaced by a global counting estimate.

The Tower Estimate

The remaining analytic input is the standard tower estimate for canonical Souza representations:

$$\begin{cases} \left(\sum_{j \geq 0} T_j(R)^q \right)^{1/q} \leq C_{\text{tow}}(\text{pqCost}(R) + M), & q < \infty, \\ \sup_{j \geq 0} T_j(R) \leq C_{\text{tow}}(\text{pqCost}(R) + M), & q = \infty. \end{cases} \quad (19)$$

This is the paper-level form of the “tower term”. It is a discrete Hardy estimate along the ancestor chains of the grid. The point of (19) is that one controls the whole sequence $(T_j(R))_j$, not merely the pointwise bound $|\mathcal{T}_R(Q)| \leq M$. The latter would introduce a false factor counting the number of cells on level j .

Mixed Cost Estimate

Let S_Γ be the representation of h_Γ obtained from the above product blocks. From (18),

$$\text{levelCoeffPower}_j(S_\Gamma) \leq 2^{p-1} N^p (T_j(R)^p + A_j(R)^p).$$

Taking p -roots gives

$$\text{levelCoeffPower}_j(S_\Gamma)^{1/p} \leq 2^{1-\frac{1}{p}} N (T_j(R)^p + A_j(R)^p)^{1/p}.$$

Since

$$(a^p + b^p)^{1/p} \leq a + b \quad (a, b \geq 0),$$

we obtain

$$\text{levelCoeffPower}_j(S_\Gamma)^{1/p} \leq 2^{1-\frac{1}{p}} N (T_j(R) + A_j(R)).$$

Taking the ℓ^q norm in j , using Minkowski's inequality and (19),

$$\begin{aligned} \text{pqCost}(S_\Gamma) &\leq 2^{1-\frac{1}{p}} N (\|(T_j(R))_j\|_{\ell^q} + \|(A_j(R))_j\|_{\ell^q}) \\ &\leq C_1 N (\text{pqCost}(R) + M), \end{aligned} \quad (20)$$

where C_1 depends only on s, p, q and the grid constants.

Pointwise and L^∞ Estimates

For a fixed point x , pairwise disjointness of the domains implies that at most one indicator $\mathbf{1}_{\Omega_i}(x)$ is nonzero. If $f(x) \neq 0$, choose an active source cell P containing x . Then every i with $\mathbf{1}_{\Omega_i}(x) \neq 0$ satisfies $P \cap \Omega_i \neq \emptyset$. Hence (1) gives

$$\sum_{i \in \Lambda} |\Theta_i \mathbf{1}_{\Omega_i}(x)| \leq N.$$

Therefore

$$|h(x)| = \left| \sum_{i \in \Lambda} \Theta_i \mathbf{1}_{\Omega_i}(x) f(x) \right| \leq N |f(x)| \leq NM$$

for a.e. x . Thus

$$\|h\|_{L^\infty} \leq NM. \quad (21)$$

Passage to the Infinite Sum

Let $(\Gamma_n)_n$ be an increasing sequence of finite subsets of Λ exhausting Λ , and set

$$h_n = \sum_{i \in \Gamma_n} \Theta_i \mathbf{1}_{\Omega_i} f.$$

The estimates (20) and (21) are uniform in n . By the compactness/limit theorem for bounded families of Besov representations, there is a subsequence of the representations S_{Γ_n} converging to a representation S of

$$h = \sum_{i \in \Lambda} \Theta_i \mathbf{1}_{\Omega_i} f$$

with

$$\text{pqCost}(S) \leq C_1 N (\text{pqCost}(R) + M).$$

The support condition is closed under this limit: if a coefficient of S on a cell Q is nonzero, then $Q \subset \Omega_i$ for some $i \in \Lambda$. Combining this with (21), and increasing the constant if necessary, gives

$$\text{pqCost}(S) + \|h\|_{L^\infty} \leq C_{\text{na}} N (\text{pqCost}(R) + M).$$

This proves (2).

Remark 1. *In the Lean development, the already formalized level estimate is exactly (18). The term*

$$T_j(R)^p = \sum_{Q \in \mathcal{P}^j} |\text{weightedAncestorCoeffSum}(G, R, Q)|^p$$

is the tower term. The remaining formal analytic step is the mixed ℓ^q estimate (19); replacing $T_j(R)$ by the pointwise bound M times the number of level cells is not the desired argument.